

Aim: Market Basket Analysis.**1.Importing libraries.**

```
import pandas as pd
import micropip
await micropip.install(['mlxtend'])
from mlxtend.frequent_patterns import apriori, association_rules
```

2.Creating a dataset.

```
dataset = [
    ['milk', 'bread', 'nuts', 'apple'],
    ['milk', 'bread', 'nuts'],
    ['milk', 'bread'],
    ['milk', 'bread', 'apple'],
    ['milk', 'bread', 'apple']
]
```

/lib/python3.12/site-packages/pyodide_kernel/kernel
future. Please pass the result to `transformed\_cell`
7.17 and above.

if self.interpreter.should_run_async(code):

3.Using TransactionEncoder for human readable format.**CODE:**

```
from mlxtend.preprocessing import TransactionEncoder
te = TransactionEncoder()
te_data = te.fit(dataset).transform(dataset)
df = pd.DataFrame(te_data, columns = te.columns_)
print("Transaction DataFrame:")
print(df)
```

OUTPUT:

```
Transaction DataFrame:
   apple  bread  milk  nuts
0   True   True   True   True
1  False   True   True   True
2  False   True   True  False
3   True   True   True  False
4   True   True   True  False
```

4.Checking how many times the items has occurred.**CODE:**

```
frequent_itemsets = apriori(df, min_support = 0.6, use_colnames = True)
print("\nFrequent Itemsets:")
print(frequent_itemsets)
```

OUTPUT:

Frequent Itemsets:

	support	itemsets
0	0.6	(apple)
1	1.0	(bread)
2	1.0	(milk)
3	0.6	(bread, apple)
4	0.6	(milk, apple)
5	1.0	(milk, bread)
6	0.6	(milk, bread, apple)

5.Checking the association of each items with each other.

CODE:

```
rules = association_rules(frequent_itemsets, metric = "confidence", min_threshold = 0.7)
print("\nAssociation Rules:")
print(rules[['antecedents', 'consequents', 'support', 'confidence', 'lift']])
```

OUTPUT:

Association Rules:

	antecedents	consequents	support	confidence	lift
0	(apple)	(bread)	0.6	1.0	1.0
1	(apple)	(milk)	0.6	1.0	1.0
2	(milk)	(bread)	1.0	1.0	1.0
3	(bread)	(milk)	1.0	1.0	1.0
4	(milk, apple)	(bread)	0.6	1.0	1.0
5	(bread, apple)	(milk)	0.6	1.0	1.0
6	(apple)	(milk, bread)	0.6	1.0	1.0